Scheduling rules

Abbreviations/lingo:
EM = Emergency Medicine
OSR = Off service rotator
PGY = post grad year, or year of residency
Intern = PGY 1
Junior = PGY 1 or PGY 2
Senior = PGY 3 or PGY 4
Golden weekend = full weekend off (saturday and sunday)
Block = 2 or 4 week period of time assigned to work in the department
BH = Bellevue hospital
TH = Tisch hospital
BK = Brooklyn
WC = wildcard shift
Junior shifts designed with lower case letters, senior shifts designated with upper case letters. (for example, d2a is a junior shift, D2a is a senior shift)

Overview

Overall numbers:
Each resident will be assigned to work 9 shifts/2 week block, or 18 shifts/4 week block
Each chief will be assigned to 7 shifts/2 week block, or 14 shifts/4 week block
If you have fully staffed the required shifts, and you have residents who have not worked the maximum (9 shifts/2 week block or 18 shifts/4 week block), you can bank these extra shifts as "owed shifts." You can additionally create additional float shifts (DF/KF) for seniors to work if there are many extra shifts.
Junior residents work 10 hour shifts during the weekdays, and 12 hour shifts on the weekends.
PGY3 residents work 10 hour shifts regardless of weekdays or weekends.
PGY4 residents work 8 hour shifts regardless of weekdays or weekends.
All Brooklyn shifts are 10 hours, regardless of PGY.

Duty Hours:
The academic week starts at 00:00 Sunday AM and runs through the following Saturday 11:59 PM
Each resident must not work more more than 60 hours per academic week.
Our EM blocks run Monday through Sunday. Even if a resident is starting an EM block on Monday, you have to include their worked hours from Sunday in counting their duty hours for the academic week.
Each resident must have at least 1 full 24 hour period off within the academic week.
Each resident must have at least the same number of hours off that they just worked, prior to their next shift. For example, if a resident works 12 hours, their next shift can not start until 12

hours after the end of the first shift. As another example, a resident can not work a night shift on Monday night which ends at 8 am, followed by a day shift on Tuesday that starts at 8 am. Residents can not work more than 12 hours straight in the Emergency Department.

Shift restrictions:
Certain shifts can only be staffed by certain residents. There are designations between junior and senior shifts. There are additionally PGY specific restrictions, such as the senior team 2 shifts at Bellevue, which must be staffed by a PGY4.
At Bellevue only, we have off-service rotators who staff shifts in the department.
All off service rotators are considered juniors and fill junior shifts.
Depending on the service they come from, they are designated as anchor or non anchor services. Ortho, Internal medicine, Plastic surgery and Ophthalmology are anchor services. Prelims, OMFS and psych are non anchor.

Site specifics:
PGY3s and 4s can work shifts at either Bellevue, Tisch or Brooklyn during an EM block.
PGY2s can work shifts at either Bellevue or Tisch
PGY1s will be assigned to a specific site, either Bellevue or Tisch. As much as possible, they should be assigned to only shifts at that site. If needed, you can mix a few shifts from the other site into their schedule.
The off service rotators do not staff shifts at Tisch or Brooklyn.

Nights:
There are no official restrictions on how many night shifts in a row a resident can work. However, try to schedule residents for no more than 6 night shifts in a row, for wellness/quality of life sake. This issue usually only arises with PGY2s, who work the heaviest burden of night shifts.

Weekends:
Residents should have off 1 full weekend off (aka golden weekend) in a 2 week block, or 2 weekends in a 4 week block. These do not need to be alternating weekends. For example, a resident can have off the first weekend and last weekend in a 4 week block, and work the two middle weekends.
Priority for golden weekends should be given to the Emergency Medicine residents. Off service rotators have less priority for having full weekends off, though in an ideal case,
If a resident has too many days requested off, this can conflict with their golden weekends. This is the only case when EM residents should not be given their allotted number of golden weekends.

Requests:
Residents can request 2 days off in a 2 week block, or 4 days off in a 4 week block.
These should be honored as much as possible.

Residents will often request more than their allotted number of days off. In this case, they should have listed their priority days off. (You can confer with the resident about their preference if not listed).
Off service rotators can request the same number of days off.
Not every resident will request days off. Scheduling should prioritize those who have made specific requests.

Adjuncts:
Residents who are assigned to AnUS, Psych, Hand, Tox can be assigned to work “adjunct” shifts in the ED.
Adjunct shifts can only be worked on the weekend. They can not be staffed to work week day shifts.
They should be avoided if possible but are sometimes necessary.
You can only assign 1 adjunct shift per resident per weekend. They can work multiple weekends. For example, a resident on AnUS can work Saturday of the first weekend, and Saturday of the second weekend, but not Saturday/Sunday of the same weekend.
Tox adjunct residents should only be scheduled for maximum of 2 weekends in their 4 week block, as this is a more intensive rotation.

Wednesdays:
Wednesday staffing models are the most unique.
No residents should be scheduled for shifts Wednesday morning, during our protected academic conferences.
On Wednesday afternoons, there will be assigned:
Bellevue
- 3 PGY 3 senior shifts D1, D3, D4. These shifts are 1p-8p on Wednesdays
- 3 junior shifts: d2b (1p-10p), d4 (2p-12a), d5 (2p-12a)
- 2 peds shifts: PT4, PB4. These shifts are 4p-12a, similar to other days.

Tisch
- 1 single junior assigned to wildcard shift. Wildcard is not team specific. It is a 1p-9p shift

Brooklyn
- There are no Wednesday shifts at Brooklyn

Float shifts:
Float shifts do not need to be scheduled.
As mentioned earlier, if you have many extra shifts, you can schedule senior residents for float shifts. These can be scheduled on any weekday.
These shifts are 8 hours for PGY4s and 10 hours for PGY3s.
Tisch Float = TF. The hours are 8a-4p.
Bellevue float shifts are named Day Float = DF. These shifts are 8a-4p, except for on Wednesdays when they are 12p-8p.

Teaching shifts:

During the months of July, August, September and October, we will have rotating medical students.
We are not required to schedule/assign the medical students their schedules. However, there should be senior residents assigned to teaching shifts (DT Bellevue teaching or TT Tisch teaching, both 8a-4p) that align with the medical students' schedules.
UME faculty will give you these schedules.

Annaliese's manual algorithm:

- Double check the off service rotators for the block
- Include all resident request days off
- Staff the most constrained shifts first (PGY2 night shifts n1/n3, any non anchor residents, examining resident shifts)
- Fill out all the nights and weekends
- Last step plug and play all the missing day shifts with other residents
- Check the counters to make sure you aren't missing any shifts
- Double check duty hour violations

SHIFT MODELS

JUNIORS

Juniors = PGY1 and PGY2
Interns = PGY1

Bellevue Notes:
All OSR are juniors, and should be considered PGY1 equivalent.
Of PGY1 shifts, there are anchor and non anchor shifts. The only non-anchor shifts are on team 2 at Bellevue. Non anchor shifts include n2b, d2b (except on Wednesday, when this must be staffed by a PGY2) and d2/n2 (weekend shifts).
PGY2 specific shifts: d2a, n1, n3 and d2b on wednesday must be staffed by a PGY2.
There must always be a PGY2 on team 2 during the day, including on the weekend, to act as examining resident. Therefore, one of the d2 weekend shifts must be staffed by a PGY2 as well.
Other junior shifts including d1, d3, d4a, d4, d5 can be staffed by any anchor juniors.

Tisch Notes:
PGY1s and 2s can work any shifts at Tisch including k2a, k2b, th1a, wc and t1n.
t1n shifts should be preferentially assigned to PGY1s.
If there is a PGY1 and PGY2 working tisch day shifts, the PGY2 should be assigned to the k2b shift and the PGY1 should be assigned to the k2a shift.

The junior staffing models for Bellevue and Tisch are explained below. One is written form, one is a table key.

Types of shifts:

- Day
    - Bellevue
        - Weekday: 2
            - 8-19
        - Weekend: 3
            - 8-20
    - Tisch
        - Weekday: 3
            - 2 @ 8-18, 1 @ 10-20
        - Weekend: 2
            - 8-20
- Swing
    - Bellevue
        - Weekday: 5
            - 3 @ 12-22, 2 @ 14-00
    - Tisch

            - Weekday: 2
                - 14-00
- Night
    - Bellevue
        - Weekday: 4
            - 1 @ 19-5, 3 @ 22-8
        - Weekend: 3
            - 20-8
    - Tisch
        - Weekday: 1
            - 20-6
        - Weekend: 1
            - 20-8

Cumulative staffing needs in one week:
Days: 10+6+15+4 = 35
Swing: 25+10=35
Nights: 20+6+5+2 = 33
Grand total of 103 shifts to be staffed each week, 206 shifts per block.
There are a total number of 5356 junior shifts to be staffed annually (there are 5508 total ED resident shifts available)

OSR/PGY1/PGY2 Schedule:
Bellevue
WEEKEND
d1: Team 1 8a - 8p
d2: Team 2 8a - 8p
d3: Team 3 8a - 8p
n1: Team 1 8p - 8a
n2: Team 2 8p - 8a
n3: Team 3 8p - 8a
WEEKDAY
d1: Team 1 12p - 10p (including Wed)
d2a: Team 2 8a - 7p (expected at LMR)
d2b: Team 2 12p - 10p
d3: Team 3 12p - 10p (including Wed)

d4a: Team 4 8a - 7p
d4: Team 4 2p - 12a
d5: Team 5 2p - 12a
n1: Team 1 10p - 8a
n2a: Team 2 7p - 5a
n2b: Team 2 10p - 8a
n3: Team 3 10p - 8a
NYU
Weekday
TH Team 1 Junior Night (T1N) = 8p - 6a
Wild card (WC) Wednesday only = 1p - 9p
Kimmel Team 2 Junior day (K2a) = Red Team 8a - 6p
Kimmel Team 2 Junior swing (K2b) = Red Team 2p-12a
Weekends
TH Team 1 Junior (TH1a) = 8a - 8p
Kimmel Team 2 Junior (K2a) = 8a - 8p
TH Team 1 Night (T1N) = 8p - 8a

SENIORS

Seniors = PGY3 and PGY4
Seniors will work at all hospital sites in their EM blocks.
They will additionally be assigned to work peds shifts in an EM block.
There should be 2 seniors assigned to peds shifts each day. They can be either PGY3 or PGY4. However, the shifts will be 8 hours for PGY4s and 10 hours for PGY3s. For example, PB2 is a PGY3 shift, which is 2p-12a, and PB4 is a PGY4 shift, which is 4p-12a.
Bellevue Team 2 seniors shifts, designated with D2a, D2p or N2, can only be worked by PGY4s. On weekdays M/T/Th/Fr, the D2a PGY4 will be assigned to give a morning report, designated by pink lettering D2a.

PGY3 Schedule:
Bellevue
Weekdays
D3: Team 3 11a - 9p (1p - 8p on Wed)
D4: Team 4 11a - 9p (1p - 8p on Wed)
D5: Team 5 11a - 9p (1p - 8p on Wed)
D5p: Team 5 2p - 12a
D1: Team 1 11a - 9p (1p - 8p on Wed)
NF: 9p - 7a (8p - 6a on Wed)
Weekends
D1a: Team 1 7a - 5p Sat, 8a-6p Sun
D3p: Team 3 12p - 10p
NF: 10p - 8a
NYU
T1a: Tisch Team 1 (Blue Team) 8a - 6p
T1p: Tisch Team 1 (Blue Team) 2p - 12a
T1N: Tisch Team 1 (Blue Team) 10p - 8a
T2a: Kimmel Team 2 (Red Team) 8a - 6p
T2p: Kimmel Team 2 (Red Team) 2p - 12a
WC: Tisch/Kimmel Float 1p - 9p
KF: Kimmel Float 12p-10p

PGY3 Peds:
PB8: PES 8a - 6p
PB2: PES 2p - 12a
PT8: PECC 8a - 6p
PT2: PECC 2p - 12a
PT/PB12: PECC/PES 12p - 10p (Tuesdays)
PT9: PECC 9a-7p
PGY3 Brooklyn:
BK9: Brooklyn 9a-7p
BK2: Brooklyn 2p-12a
BKA: Brooklyn A-side 10a-8p
BKN: Brooklyn night 8p-6a

PGY4 Schedule:
Bellevue
D1: Team 1 1pm-9pm
D2a: Team 2 8a - 4p
D2a (pink): morning report
D2p: Team 2 4p - 12a
N2: Team 2 1145p - 8a
DT: 8a-4p Bellevue Teaching Shift
DF: Day Float 8a-4p
DF: Day Float 12p-8p (Wednesday)
NYU
T1a: Tisch Team 1 (Blue Team) 8a - 4p
T1p: Tisch Team 1 (Blue Team) 4p - 12a
T1N: Tisch Team 1 (Blue Team) 12a - 8a
T2a: Kimmel Team 2 (Red Team) 8a - 4p
T2p: Kimmel Team 2 (Red Team) 4p - 12a
WC: Tisch/Kimmel Float 1p - 9p
TT: Tisch Teaching Shift 8a-4p

TF: Tisch Float 8a - 4p
KF: Kimmel Float 12p-8p
PGY4 Peds:
PB8: PES 8a - 4p
PT/PB12: PECC/PES 12p - 8p (Tuesdays)
PB2: PES 2p - 12a (weekend)
PB4: PES 4p - 12a
PT8: PECC 8a - 4p
PT2: PECC 2p - 12a (weekend)
PT4: PECC 4p - 12a
PT9: PECC 9a-5p
PT`10: PECC 10a-6p
PT11: PECC 11a-7p
PGY4 Brooklyn:
BK9: Brooklyn 9a-7p
BK2: Brooklyn 2p-12a
BKA: Brooklyn A-side 10a-8p
BKN: Brooklyn night 8p-6a

Problem:
We are trying to figure out how to optimally satisfy the two hospitals' need for resident labor for the junior resident cohort. We have X number of shifts that need to be staffed by residents in every 2 week block. We only have a limited number of residents available during each block and they should be scheduled for an equal number of shifts. Residents will request certain days off during each block. We can not create schedules that cause duty hour violations.
Constraints

- 1 day off in 7 days for each calendar week (calendar week starts Sunday)
- For a given shift, you must have have the same number of hours off afterwards as you worked during the shift
- You work 18 shifts out of 28 days
- Total shifts:
- You can only schedule the residents in an ED block for shifts
- The interns must stay at one designated hospital site for the duration of the 2 week block, whereas PGY2s can work at either TH or BH in any 2 week block
    -

Penalties

- Unscheduled shifts
    - Conflicts with requests

Other rewards

- Protecting Tuesday night off if possible
    - Schedule off service resident on team 5 swing

Blocks :
What is a block?
A block is a two week period of time. A block designates where you are working for that two week period of time. Throughout the year, residents spend time in other specialties outside the ER aka "off service". We can only staff residents who are on an ED block at the time.

- Each block is 2 weeks
- There are 26 2 week blocks in one year
- The interns have orientation for the first 2 blocks, and can not be scheduled for ED shifts
- There are 18 residents in each class

Terminology:
When I refer to residents in this document, I am referring to juniors, consider these two words interchangeable. When I use the term resident or juniors, I am referring to both of the PGY1s and PGY2s, a total of 36 residents. If I want to refer specifically to one class, I will specify by either saying intern/PGY1 or PGY2
Juniors = PGY1/intern (18) + PGY2 (18)
BH = Bellevue hospital
TH= tisch hospital
PGY1 blocks (number in parentheses is the number of this block annually)

- BH (2) + TH (4)
    - X 18 residents = 108 blocks = 1944 shifts
- BK (2)
- Peds TH (1)
- Peds BH (1)
- Medicine (2)
- Micu (2)
- SICU (2)
- Trauma (2)
- Nicu (1)
- Cpep (1)
- Jeopardy (1)
- Anus (1)
- Vacation (2)

PGY2 year blocks

- BH/TH (11)
    - X 18 residents = 198 blocks = 3564 shifts
- BK (3)
- Peds BH (1)
- Peds TH (1)
- ortho (2)
- Anus (1)
- OB (1)
- CCU (2)
- Jeopardy (1)
- EMS (1)
- Vacation (2)

Types of shifts:

- Day
    - Bellevue
        - Weekday: 2
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        - Weekend: 3
            - 8-20
    - Tisch
        - Weekday: 3
            - 2 @ 8-18, 1 @ 10-20
        - Weekend: 2
            - 8-20
- Swing
    - Bellevue
        - Weekday: 5
            - 3 @ 12-22, 2 @ 14-00
    - Tisch
        - Weekday: 2

            - 14-00
- Night
    - Bellevue
        - Weekday: 4
            - 1 @ 19-5, 3 @ 22-8
        - Weekend: 3
            - 20-8
    - Tisch
        - Weekday: 1
            - 20-6
        - Weekend: 1
            - 20-8

Cumulative staffing needs in one week:
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Swing: 25+10=35
Nights: 20+6+5+2 = 33
Grand total of 103 shifts to be staffed each week, 206 shifts per block.
There are a total number of 5356 junior shifts to be staffed annually (there are 5508 total ED resident shifts available)
Week days are considered Monday - Friday, weekends are considered Saturday and Sunday.
Weekday shifts are 10 hours long. Weekend shifts are 12 hours long.
Due to conference on Wednesday morning, no residents need to staff the day time shifts. The swing and night shift coverage on Wednesday is otherwise the same.
Other Bellevue specifics:
Anchor vs non anchor
What is an anchor? An anchor is a resident from the following specialties: EM, ortho, internal medicine, plastics
Non anchor residents are residents from the following specialties: psych, OMFS, ophtho
Most ED shifts are staffed by anchors. The only non anchor shifts are on team 2, one is a swing shift, and one is a night shift.
Pgy2 specifics
There must be a pgy2 on team 2 day shift.
There must be 2 PGY2s every night shift at Bellevue, one on team 1 and one on team 3.
What is a team? A team is one specific group of on shift providers taking care of the same group of patients. Each team has an assigned location and is overseen by a single attending. A team consists of a mix of junior and senior residents, depending on the team/day. There are 5 teams to staff on week days. There are 3 teams to staff on nights and weekends (team 4 and 5 don't exist at these times).
What are the requirements, if any, for mix of residents on a team? Operations designates the arrangement of senior to juniors on the teams. The senior residents are scheduled independently of the junior residents, though will work alongside them while on shift.
What are the requirements for team size and anchor?
The team size depends on the time of day. There is always one attending per team. During less busy hours, there can be as few as one resident working with the attending, for a total team size of two. The team can be as large as attending + 1 senior + 2 juniors for a total team size of four. There is almost never more than one senior.

Bellevue staffing (unless otherwise listed, all anchor shifts). The below outline shows the team based staffing requirements for juniors at Bellevue, for each day of the week.

Team 1

- Monday:
  - 1 swing
  - 1 night
- Tuesday:
  - 1 swing
  - 1 night
- Wednesday:
  - 1 night
- Thursday:
  - 1 swing
  - 1 night
- Friday:
  - 1 swing
  - 1 night
- Saturday:
  - 1 day
  - 1 night
- Sunday:
  - 1 day
  - 1 night

Team 2

- Monday:
  - 1 day
  - 1 swing NON ANCHOR
  - 1 night
  - 1 night NON ANCHOR
- Tuesday:
  - 1 day
  - 1 swing NON ANCHOR
  - 1 night
  - 1 night NON ANCHOR
- Wednesday:
  - 1 swing NON ANCHOR
  - 1 night
  - 1 night NON ANCHOR
- Thursday:
  - 1 day
  - 1 swing NON ANCHOR
  - 1 night
  - 1 night NON ANCHOR
- Friday:
  - 1 day
  - 1 swing NON ANCHOR

        - 1 night
        - 1 night NON ANCHOR
    - Saturday:
        - 1 day
        - 1 day NON ANCHOR
        - 1 night
        - 1 night NON ANCHOR
    - Sunday:
        - 1 day
        - 1 day NON ANCHOR
        - 1 night
        - 1 night NON ANCHOR

Team 3

- Monday:
    - 1 swing
    - 1 night
- Tuesday:
    - 1 swing
    - 1 night
- Wednesday:
    - 1 night
- Thursday:
    - 1 swing
    - 1 night
- Friday:
    - 1 swing
    - 1 night
- Saturday:
    - 1 day
    - 1 night
- Sunday:
    - 1 day
    - 1 night

Team 4 - WEEKDAY DAYTIME ONLY

- Monday:
    - 1 day
    - 1 swing
- Tuesday:
    - 1 day
    - 1 swing
- Wednesday:
    - 1 swing
- Thursday:
    - 1 swing
- Friday:

        - 1 swing

Team 5 - WEEKDAY DAYTIME ONLY

- Monday:
    - 1 swing
- Tuesday:
    - 1 swing
- Wednesday:
    - 1 swing
- Thursday:
    - 1 swing
- Friday:
    - 1 swing

The following is the team based staffing requirements for juniors at tisch hospital.
Team 1/blue

- Monday:
    - 1 night
- Tuesday:
    - 1 night
- Wednesday:
    - 1 night
- Thursday:
    - 1 night
- Friday:
    - 1 night
- Saturday
    - 1 day
    - 1 night
- Sunday
    - 1 day
    - 1 night

Team 2/red

- Monday:
    - 1 day
    - 1 swing
- Tuesday:
    - 1 day
    - 1 swing
- Wednesday:
    - None
- Thursday:
    - 1 day
    - 1 swing
- Friday:

        - 1 day
        - 1 swing
- Saturday:
    - 1 day
- Sunday:
    - 1 day

PLUS ONE WILD CARD SHIFT

- One lone Wednesday afternoon shift scheduled at tisch = wild card, not team specific
- Concept of the wild card shift: this is a Wednesday specific post conference shortened shift. Operations can flex this resident to whichever team they are needed, so I will not assign them a team myself, simply schedule an additional “wildcard” resident

Annual shift breakdown per resident
Days: 35

- 1820 day shifts annually
- =

Swing: 35

- 1820 swing shifts annually
- = 50.5 swing shifts/resident per year - need to clarify w operations if interns are restricted to non swing shifts

Nights: 33

- 33 night shifts/week = 1716 night shifts annually = 30% interns and 70% PGY2s
- 515 intern night shifts = 28.6 night shifts per intern per year
    - =4.8 night shifts/block
- 1201 pgy2 night shifts = 66.7 night shifts per pgy2 per year
    - =6 night shifts/block

Weekends: 18

- 18 weekend shifts/week= 936 weekend shifts annually
- = 26 weekend shifts/resident per year